HW #7

ALGEBRAIC NUMBER THEORY, GU4043; INSTRUCTOR: GYUJIN OH

Reading Homework. Try Exercises A.6, A.7, A.8 in the textbook. Read its solutions in the back.

Question 1. (Do not use §4.3-4.5!)

(1) Let A be a Dedekind domain. Show that there is a one-to-one correspondence

$$\{\text{Nonzero prime ideals of } A\} \leftrightarrow \left\{ \begin{array}{c} \text{Normalized discrete valuations } v \text{ on} \\ \text{Frac}(A) \text{ such that } v(x) \geq 0 \text{ for all } x \in A \end{array} \right\}$$
$$\mathfrak{p} \subset A \mapsto v_{\mathfrak{p}} \text{ induced by the DVR } A_{\mathfrak{p}}$$
$$\{x \in A \,:\, v(x) \geq 1\} \leftarrow\!\shortmid v.$$

(2) Let L/K be an extension of number fields, and let $\mathfrak{p}$ be a nonzero prime ideal of $\mathcal{O}_K$. Show that the one-to-one correspondence of (1) for $A = \mathcal{O}_L$ restricts to a one-to-one correspondence

$$\{\text{Prime ideals } \mathfrak{q} \text{ of } \mathcal{O}_L \text{ dividing } \mathfrak{p}\} \leftrightarrow \left\{ \begin{array}{c} \text{Normalized discrete valuations } v \text{ on } L \text{ such that} \\ v|_K = ev_{\mathfrak{p}} \text{ for some } e \in \mathbb{Z}_{\geq 1} \end{array} \right\}.$$

Here, $v|_K$ is the restriction of $v : L \to \mathbb{Z}_{\geq 0} \cup \{\infty\}$ to K (i.e., the composition of v with the inclusion $K \hookrightarrow L$).

(3) In the situation of (2), if $\mathfrak{q}$ is a prime ideal of $\mathcal{O}_L$ dividing $\mathfrak{p}$, show that $v_{\mathfrak{q}} = e(\mathfrak{q}|\mathfrak{p})v_{\mathfrak{p}}$.

Question 2. Let K be a number field, and let $\alpha \in K^\times$ that is not a square in K. Consider $L = K(\sqrt{\alpha})$, which is a quadratic extension of K.

(1) Let $\mathfrak{p}$ be a nonzero prime ideal of $\mathcal{O}_K$. If $v_{\mathfrak{p}}(\alpha)$ is odd, show that $\mathfrak{p}$ is ramified in L.

Hint. Use Question 1.

(2) Let $\mathfrak{p}$ be a nonzero prime ideal of $\mathcal{O}_K$ that does not divide 2. If $v_{\mathfrak{p}}(\alpha)$ is even, show that $\mathfrak{p}$ is unramified in L.

Hint. Take a uniformizer of $\mathcal{O}_{K,\mathfrak{p}}$ to reduce to the case of $v_{\mathfrak{p}}(\alpha) = 0$, and use the relative discriminant.

(3) Find an example of K, α, and $\mathfrak{p}$ (which necessarily divides 2), such that $\mathfrak{p}$ is ramified in L, even though $v_{\mathfrak{p}}(\alpha)$ is even.

(4) Let $\mathfrak{p}$ be a prime ideal of $\mathcal{O}_K$ that divides 2. If α is a square modulo $\mathfrak{p}^{2e(\mathfrak{p}|2)}$ (namely, if there exists $x \in \mathcal{O}_K$ such that $x^2 \equiv \alpha \pmod{\mathfrak{p}^{2e(\mathfrak{p}|2)}}$), show that $\mathfrak{p}$ is unramified in L.

Hint. Take a uniformizer of $\mathcal{O}_{K,\mathfrak{p}}$ to reduce to the case of $v_{\mathfrak{p}}(\alpha) = 0$. For x as in the problem, show that $\frac{x+\sqrt{\alpha}}{2} \in \mathcal{O}_{K(\sqrt{\alpha}),\mathfrak{q}}$ for any prime $\mathfrak{q}$ of $\mathcal{O}_{K(\sqrt{\alpha})}$ dividing $\mathfrak{p}$.

Question 3. Later in the course, when we learn class field theory, we will see that, for any number field K, there is a very special abelian extension H_K/K, called the **Hilbert class field of** K, such that $\mathrm{Gal}(H_K/K) \cong \mathrm{Cl}(K)$ and every prime ideal of K is unramified in H_K.

Although the above general fact is currently beyond our reach, in this question, we will show a very particular consequence of this general fact by hand: **for an imaginary quadratic field K with h_K even, there exists a quadratic extension L/K in which every prime ideal of K is unramified**.

Let $K = \mathbb{Q}(\sqrt{-n})$ be an imaginary quadratic field, with n a squarefree positive integer, such that h_K is even.

(1) Show that $\mathrm{disc}(K)$ has at least two prime factors.

Hint. Use Exercise 3.4.

(2) Use Question 2 to show that, if $m \neq 1$ is any odd divisor of n, then either $K(\sqrt{m})$ or $K(\sqrt{-m})$ is a desired quadratic extension (i.e., where all prime ideals of K are unramified).

Hint. For prime ideals of K not dividing 2, use that fact that all rational prime factors of n are ramified in K. For prime ideals of K dividing 2, use the fact that either m or $-m$ is $1 \pmod 4$. Remember that such a quadratic extension exists if and only if h_K is even, so your proof must use this (e.g., Exercise 3.4) at some point.